Sequential Conditional Means of Equicorrelated Scores

One, two or three fixed starting scores included in the average

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Each test has population mean 0 and standard deviation 1. Fix the first observed score at a. For jointly Gaussian tests with equal pairwise correlation c, recursively predict each subsequent test by conditioning on the starting score and on all earlier predictions. Every subsequent prediction equals ca. Including a as the first value, the average of $a, e_1, \dots, e_k$ is $a(1 + kc)/(k + 1)$. This fixed recursive average has standard deviation zero. The actual $(k+1)$-test average has population standard deviation $\sqrt{(1 + kc)/(k + 1)}$; after observing the starting score, its conditional standard deviation is $\sqrt{k(1 - c)(1 + kc)}/(k + 1)$. We derive the Gaussian conditional mean, solve the recursion, describe the admissible correlations and singular endpoints, and distinguish these sources of variation. A general derivation also gives the cases with two or three initial scores all fixed at a, including every starting score in the average. Worked examples use initial IQ scores of 150 on a mean-100, SD-15 reporting scale, correlation 0.6, and one, two or ten subsequent predictions. A counterexample shows that means, variances and correlations alone do not determine the recursive conditional means. Every mathematical conclusion is derived from explicitly stated reference theorems; exact arithmetic certificates justify the displayed numerical rounding.

AI disclosure. This analysis, its mathematical exposition, supporting repository documentation and typesetting were prepared with OpenAI Codex and collaborating AI agents. The requester specified the problem, clarified the indexing and requested this analysis. Mathematical review and build checks were performed within the same AI-assisted workflow; no independent human peer review or new empirical psychometric validation is claimed.

Proof standard. Model assumptions and reporting-scale definitions are stated explicitly. The mathematical conclusions are proved from the established results collected in Section 2. Section 12 maps the claims to their proofs, and Section 11 supplies exact numerical certificates. The references identify the versions and theorem locators actually checked; mathematical deductions do not establish the empirical adequacy of the chosen test model.

1 Problem and notation

Let $k \geq 1$ be an integer and write

$$X_0, X_1, \dots, X_k$$

for standardized test scores. Throughout this analysis, every test has population mean zero and standard deviation one:

$$\mathrm{E}[X_j] = 0, \qquad \mathrm{SD}(X_j) = 1, \qquad \mathrm{Var}(X_j) = 1, \qquad 0 \leq j \leq k. \tag{1}$$

These parameters describe the reference distribution of each test; a fixed observed score a need not equal zero. We relabel the original tests $A_r(c, 0, 1)$ so that test 0 supplies the starting observation. For distinct indices, assume $\mathrm{Cov}(X_i, X_j) = \mathrm{Corr}(X_i, X_j) = c$.

Use $a \in \mathbb{R}$ for the *fixed starting score*; it is a scalar, not a function. For each $1 \le n \le k$, let g_n be the Gaussian conditional mean function of X_n given $(X_0, \dots, X_{n-1})$. Define

$$e_1 = g_1(a) = \mathrm{E}[X_1 \mid X_0 = a], \tag{2}$$

$$\begin{aligned} e_n &= g_n(a, e_1, \dots, e_{n-1}) \\ &= \mathrm{E}[X_n \mid X_0 = a, X_1 = e_1, \dots, X_{n-1} = e_{n-1}], \qquad 2 \le n \le k. \end{aligned} \tag{3}$$

The quantity of interest is

$$\boxed{B_k = \frac{a + \sum_{n=1}^{k} e_n}{k+1}.} \tag{4}$$

It contains $k+1$ values: the starting score a is the first value, followed by the k predictions $e_1, \dots, e_k$. Thus k counts subsequent predictions, while $k+1$ counts all values in the average. Section 7 extends this setup to r initial scores all fixed at a, followed by k predictions, for $r + k$ total values.

For continuous distributions, conditioning on a particular score is not a ratio of probabilities of point events. A conditional mean is ordinarily defined only almost everywhere. Here, g_n means the continuous version supplied by Gaussian regression, derived below. This convention specifies the recursion even at individual histories of probability zero. The singular case $c = 1$ will be treated directly on its support. Existence and almost-sure uniqueness of conditional expectation are part of Theorem 2.2; the pointwise continuous-version convention is justified by Lemma 2.6.

2 Reference theorems and proof conventions

The probability model, its numerical parameters and the reporting scale are assumptions or definitions. The conclusions below are mathematical consequences of those choices. We use the cited standard foundational theorems; the proofs here establish their particular forms needed by this report. In particular, the cited Radon–Nikodym, spectral and characteristic-function uniqueness theorems are not claims of new results. All random variables in a variance or covariance calculation have finite second moments. Conditional-expectation equalities hold almost surely unless a specified continuous version is being evaluated at an individual history. Vector conditional expectations are taken coordinate by coordinate.

Theorem 2.1 (Moment identities and zero variance)**.** *For integrable random variables, expectation is linear. If $X_1, \dots, X_n$ have finite second moments, covariance is bilinear and*

$$\mathrm{Var}\left(\sum_{j=1}^{n} a_j X_j\right) = \sum_{i=1}^{n} \sum_{j=1}^{n} a_i a_j \,\mathrm{Cov}(X_i, X_j).$$

For a random vector X, a constant matrix A, and a constant vector b,

$$\mathrm{E}[AX + b] = A\,\mathrm{E}[X] + b, \qquad \mathrm{Cov}(AX + b) = A\,\mathrm{Cov}(X)A^{\mathsf{T}}.$$

In particular, $\mathrm{SD}(u + vX) = |v|\,\mathrm{SD}(X)$. *Moreover,* $\mathrm{Var}(X) = 0$ *if and only if* $X = \mathrm{E}[X]$ *almost surely.*

Proof. Linearity is the integral-linearity theorem [3, Theorem 1.6.1]. Products of centered square-integrable variables are integrable by Hölder's inequality [3, Theorem 1.6.3]. Expanding $\mathrm{E}[(X - \mathrm{E}[X])(Y - \mathrm{E}[Y])]$, distributing finite sums and applying linearity gives covariance bilinearity and the displayed scalar and matrix identities. The SD identity follows by taking the nonnegative square root of $\mathrm{Var}(u + vX) = v^2 \mathrm{Var}(X)$; see also the covariance summary in [2, Section 4.2]. If the variance is zero, Chebyshev's inequality gives

$$P\{|X - \mathrm{E}[X]| \geq \varepsilon\} \leq \frac{\mathrm{Var}(X)}{\varepsilon^2} = 0 \qquad (\varepsilon > 0)$$

[3, Theorem 1.6.4]. Taking the union of these zero-probability events for $\varepsilon = 1, 1/2, 1/3, \ldots$ proves the assertion. Its converse follows directly from the variance definition. □

Theorem 2.2 (Conditional expectation and total variance)**.** *Let $\mathcal{G}$ be a sub-σ-algebra of the probability space and let X be integrable. There exists a $\mathcal{G}$-measurable integrable variable $Z = \mathrm{E}[X \mid \mathcal{G}]$, unique up to almost-sure equality, such that*

$$\mathrm{E}[\mathbf{1}_A Z] = \mathrm{E}[\mathbf{1}_A X] \qquad \text{for every } A \in \mathcal{G}.$$

Conditional expectation is linear, fixes integrable $\mathcal{G}$-measurable variables, and equals $\mathrm{E}[X]$ when X is independent of $\mathcal{G}$. If $\mathcal{G} \subseteq \mathcal{H}$, its tower property is

$$\mathrm{E}[\mathrm{E}[X \mid \mathcal{H}] \mid \mathcal{G}] = \mathrm{E}[X \mid \mathcal{G}].$$

If H is $\mathcal{G}$-measurable and X, HX are integrable, the pull-out identity is $\mathrm{E}[HX \mid \mathcal{G}] = H\,\mathrm{E}[X \mid \mathcal{G}]$. For square-integrable X,

$$\mathrm{Var}(X) = \mathrm{Var}(\mathrm{E}[X \mid \mathcal{G}]) + \mathrm{E}[\mathrm{Var}(X \mid \mathcal{G})],$$

where $\mathrm{Var}(X \mid \mathcal{G}) = \mathrm{E}[(X - \mathrm{E}[X \mid \mathcal{G}])^2 \mid \mathcal{G}]$.

Proof. Existence follows by applying the Radon–Nikodym theorem separately to the finite positive measures $A \mapsto \mathrm{E}[\mathbf{1}_A X^+]$ and $A \mapsto \mathrm{E}[\mathbf{1}_A X^-]$ on $\mathcal{G}$. Both are absolutely continuous with respect to the probability measure restricted to $\mathcal{G}$. Subtracting their derivatives gives the defining integral identity; uniqueness gives almost-sure uniqueness. This is the construction in [3, Section 4.1 and Theorem A.4.8]. Linearity and the known-variable property follow by substituting candidates in that identity. Independence gives $\mathrm{E}[\mathbf{1}_A X] = P(A)\,\mathrm{E}[X]$, proving the independent-variable case. For the tower property, integrate over $A \in \mathcal{G} \subseteq \mathcal{H}$ twice and invoke uniqueness. These properties are respectively recorded in [3, Examples 4.1.3–4.1.4 and Theorems 4.1.9(a), 4.1.13]. The pull-out property follows first for simple H, then by approximation and truncation under the stated integrability assumptions [3, Theorem 4.1.14].

For total variance put $Z = \mathrm{E}[X \mid \mathcal{G}]$ and $R = X - Z$. The conditional-expectation projection theorem gives $Z \in L^2$ and $\mathrm{E}[RH] = 0$ for every $\mathcal{G}$-measurable $H \in L^2$ [3, Theorem 4.1.15]. Apply it to $H = Z - \mathrm{E}[X]$ and expand the square in $X - \mathrm{E}[X] = R + (Z - \mathrm{E}[X])$. The cross term vanishes, while the tower property gives $\mathrm{E}[R^2] = \mathrm{E}[\mathrm{E}[R^2 \mid \mathcal{G}]]$. This proves total variance, also stated in [2, Section 4.3] and posed in [3, Exercise 4.1.7]. □

Theorem 2.3 (Spectral criterion and Gaussian covariance construction)**.** *A real symmetric matrix S is positive semidefinite if and only if all its eigenvalues are nonnegative, and positive definite if and only if all are positive. Every positive-semidefinite matrix is a covariance matrix of a possibly singular Gaussian vector. Every covariance matrix is positive semidefinite.*

Proof. The real spectral theorem gives $S = U \Lambda U^{\mathsf{T}}$ with U orthogonal and Λ real diagonal [1, Theorem 7.29]. Hence $t^{\mathsf{T}} S t = \sum_j \lambda_j (U^{\mathsf{T}} t)_j^2$, proving both eigenvalue criteria. For nonnegative eigenvalues

let $S^{1/2} = U\,\mathrm{diag}(\sqrt{\lambda_j})U^{\mathsf{T}}$, the positive square root [1, Theorem 7.38]. If G has independent standard-normal coordinates, the vector $S^{1/2}G$ has covariance S by Theorem 2.1; this construction includes zero eigenvalues [4, Theorems 3.4.4–3.4.5, author-hosted draft]. Conversely, $t^{\mathsf{T}}\,\mathrm{Cov}(X)t = \mathrm{Var}(t^{\mathsf{T}}X) \geq 0$ for every t. □

Theorem 2.4 (Gaussian transformations and block independence)**.** *An affine transformation of a jointly Gaussian vector is jointly Gaussian, including when its covariance is singular. Its mean and covariance determine its law. If two jointly Gaussian blocks have zero cross-covariance, they are independent, even if either block is singular or constant.*

Proof. Write a Gaussian vector as $X = \mu + AG$, where G has independent standard-normal coordinates; A need not be invertible. Then $BX + b = (B\mu + b) + BAG$, proving closure [4, Definition 3.3.1 and Theorem 3.3.1, author-hosted draft]. The one-dimensional Gaussian characteristic function and independence of the coordinates of G give

$$\phi_X(t) = \mathrm{E}[e^{\mathrm{i}t^{\mathsf{T}}X}] = \exp\Big(\mathrm{i}t^{\mathsf{T}}\mu - \tfrac{1}{2}t^{\mathsf{T}}\Sigma t\Big), \qquad \Sigma = AA^{\mathsf{T}},$$

as in [4, Theorem 3.3.2 and equations (3.14), (3.18), author-hosted draft]. The uniqueness consequence of multivariate characteristic-function inversion [3, Theorem 3.10.4] shows that μ, Σ determine the distribution. For blocks (U, V) with $\mathrm{Cov}(U, V) = 0$, the displayed formula factors as $\phi_{(U,V)}(s, t) = \phi_U(s)\phi_V(t)$. By the product-measure construction and Fubini's theorem [3, Theorems 1.7.1–1.7.2], this is the characteristic function of the product of the two marginal laws. Uniqueness therefore proves independence. This proof uses no density and no covariance inverse, so it includes singular blocks. □

Theorem 2.5 (Gaussian regression with a nonsingular conditioning block)**.** *Let (U, V) be jointly Gaussian with means μ_U, μ_V and covariance blocks $A = \mathrm{Cov}(U)$, $B = \mathrm{Cov}(V)$, $C = \mathrm{Cov}(U, V)$. Suppose only that B is positive definite. A conditional distribution of U given $V = v$ is the possibly singular normal law with*

$$m(v) = \mu_U + CB^{-1}(v - \mu_V), \qquad K = A - CB^{-1}C^{\mathsf{T}}.$$

In particular its conditional mean is $m(v)$ and its conditional covariance is K. The full joint covariance need not be nonsingular.

Proof. Set $W = U - \mu_U - CB^{-1}(V - \mu_V)$. The moment identities yield

$$\mathrm{E}[W] = 0, \qquad \mathrm{Cov}(W, V) = C - CB^{-1}B = 0, \qquad \mathrm{Cov}(W) = A - CB^{-1}C^{\mathsf{T}}.$$

Theorem 2.4 shows that W is Gaussian and independent of V. Thus $U = m(V) + W$. Independence implies, for every bounded Borel function h, that a version of $\mathrm{E}[h(U) \mid V]$ is $v \mapsto \int h(m(v) + w)\, P_W(dw)$. Independence and Fubini's theorem [3, Theorem 1.7.2] verify its defining conditional integral identity. This conditional probability kernel has the stated normal law, mean and covariance. The argument also works when K has zero eigenvalues. The familiar full-rank result is given in [4, Theorems 3.5.1–3.5.2 and equations (3.44)–(3.45), author-hosted draft]. Those cited statements assume a nonsingular full joint covariance; the residual proof here supplies the version needed when only the conditioning block is nonsingular. □

Lemma 2.6 (Unique continuous conditional-mean version)**.** *If V is a Gaussian vector with positive-definite covariance, two continuous functions representing the same conditional mean given V agree at every point. In particular, the affine mean function in Theorem 2.5 is its unique continuous version.*

Proof. The nonsingular Gaussian density theorem [4, Theorem 3.4.6 and equation (3.22), author-hosted draft] gives, in dimension q,

$$f_V(v) = \frac{\exp[-\frac{1}{2}(v-\mu_V)^{\mathsf{T}}B^{-1}(v-\mu_V)]}{(2\pi)^{q/2}\sqrt{\det B}} > 0 \qquad \text{for every } v \in \mathbb{R}^q.$$

Theorem 2.2 makes any two conditional-mean versions equal P_V-almost everywhere. If continuous versions differed at one point, one coordinate of their difference would remain nonzero on some open ball. Such a ball has positive probability under the displayed density, contradicting almost-sure equality. This justifies evaluating the specified continuous version at a point history; it does not turn a probability-zero event into a probability ratio. □

Theorem 2.7 (Mathematical induction)**.** *For an assertion $P(n)$ on integers $n \geq n_0$, the base assertion $P(n_0)$ and the implications $P(n) \Rightarrow P(n+1)$ for every $n \geq n_0$ establish $P(n)$ for all such integers* [5, *Section 10.1*].

Proof. If there were a counterexample, the well-ordering principle would supply the least one. It could not be n_0, by the base assertion. Its predecessor would satisfy P, and the induction step would then rule out this least counterexample. This is a contradiction. □

3 The joint model and its correlation domain

Proposition 3.1 (Admissible correlations and Gaussian realization)**.** *For $N = k+1 \geq 2$, a centered, unit-variance, equicorrelated Gaussian vector exists exactly when $-1/k \leq c \leq 1$. Admissibility for every finite battery size is equivalent to $0 \leq c \leq 1$.*

Proof. Assume $(X_0, \dots, X_k)$ is jointly Gaussian, allowing degenerate Gaussian distributions at the endpoints. Its covariance matrix is

$$R_{k+1} = (1-c)I_{k+1} + c\mathbf{1}_{k+1}\mathbf{1}_{k+1}^{\mathsf{T}}, \tag{5}$$

where I_d is the d-dimensional identity and $\mathbf{1}_d$ is a column of d ones. Vectors orthogonal to $\mathbf{1}_{k+1}$ have eigenvalue $1-c$; the vector $\mathbf{1}_{k+1}$ has eigenvalue $1+kc$. Thus a covariance matrix for these $k+1$ scores exists exactly when

$$\boxed{-\frac{1}{k} \leq c \leq 1.} \tag{6}$$

Necessity also follows directly from $t^{\mathsf{T}}R_{k+1}t = \operatorname{Var}(t^{\mathsf{T}}X) \geq 0$, by Theorem 2.1. For sufficiency, the real spectral theorem and positive-square-root theorem give $R_{k+1} = U\Lambda U^{\mathsf{T}}$ with $\Lambda \geq 0$. If Z has independent standard-normal coordinates, then $X = U\Lambda^{1/2}Z$ is Gaussian with covariance R_{k+1}, including singular endpoints. This is Theorem 2.3 together with Theorem 2.4; see Axler [1, Theorems 7.29 and 7.38] and Gallager [4, author-hosted draft, Theorems 3.4.4–3.4.5]. If $c < 0$, sufficiently large k makes $1+kc < 0$. Conversely, $0 \leq c \leq 1$ satisfies the eigenvalue conditions for every finite k.

For $0 \leq c \leq 1$, a construction consistent with the hypothetical common-factor illustration in the repository's Hoeflin/Ferguson analysis, Appendix A, is

$$X_j = \sqrt{c}\,G + \sqrt{1-c}\,\varepsilon_j, \tag{7}$$

where $G, \varepsilon_0, \dots, \varepsilon_k$ are independent standard-normal variables. The covariance-matrix derivation below also covers the admissible negative correlations. Independence and Theorem 2.1 give $\mathrm{E}[X_j] = 0$, $\operatorname{Var}(X_j) = c + (1-c) = 1$, and, for $i \neq j$, $\operatorname{Cov}(X_i, X_j) = c$. Gaussian closure follows from Theorem 2.4, so this verifies the stated construction. □

Joint Gaussianity and the choice of population parameters remain model assumptions for real test batteries.

4 Derivation of the conditional mean

Theorem 4.1 (Equicorrelated Gaussian conditioning)**.** *For $-1/k \le c < 1$ and $1 \le n \le k$, put $d_n = 1 + (n-1)c$. The continuous conditional-mean function is $g_n(y) = c\sum_{j=0}^{n-1} y_j/d_n$, and the conditional variance is $(1-c)(1+nc)/d_n$. These formulas also hold when the full vector is singular at $c = -1/k$.*

Proof. Fix $1 \le n \le k$, initially with $c < 1$, and put

$$Y_n = (X_0, \dots, X_{n-1})^\top, \qquad R_n = \operatorname{Cov}(Y_n) = (1-c)I_n + c\mathbf{1}_n\mathbf{1}_n^\top.$$

The eigenvalue of R_n in the direction $\mathbf{1}_n$ is

$$d_n = 1 + (n-1)c.$$

It is positive throughout $-1/k \le c < 1$: for negative c, $d_n \ge 1 - (n-1)/k \ge 1/k > 0$. The other eigenvalues are $1 - c > 0$. Consequently every conditioning block is nonsingular, even if the full $(k+1)$-score distribution is singular at $c = -1/k$.

An explicit inverse is

$$R_n^{-1} = \frac{1}{1-c}\left(I_n - \frac{c}{1+(n-1)c}\mathbf{1}_n\mathbf{1}_n^\top\right). \tag{8}$$

To prove the inverse identity, let $J_n = \mathbf{1}_n\mathbf{1}_n^\top$. Then $J_n^2 = nJ_n$ and $R_nJ_n = d_nJ_n$, whence

$$R_n\frac{I_n - (c/d_n)J_n}{1-c} = \frac{R_n - cJ_n}{1-c} = I_n.$$

The factors commute because both are polynomials in J_n, so the reverse product is also I_n. This proves (8) without an unproved matrix-inversion step. Since $\operatorname{Cov}(X_n, Y_n) = c\mathbf{1}_n^\top$, the regression coefficient is

$$\operatorname{Cov}(X_n, Y_n)R_n^{-1} = \frac{c}{1+(n-1)c}\mathbf{1}_n^\top. \tag{9}$$

To establish why this coefficient gives the conditional mean, define the residual

$$W_n = X_n - \frac{c}{1+(n-1)c}\sum_{j=0}^{n-1} X_j. \tag{10}$$

For each $\ell \in \{0, \dots, n-1\}$,

$$\begin{aligned}\operatorname{Cov}(W_n, X_\ell) &= c - \frac{c}{1+(n-1)c}\sum_{j=0}^{n-1}\operatorname{Cov}(X_j, X_\ell)\\ &= c - \frac{c}{1+(n-1)c}\big[1+(n-1)c\big] = 0.\end{aligned}$$

The residual and Y_n are jointly Gaussian because they are linear transformations of a Gaussian vector, by Theorem 2.4. Their zero cross-covariance implies independence: the Gaussian characteristic function $\exp(-t^\top\Sigma t/2)$ factors across covariance blocks when the cross-covariance block is zero. This argument

also allows a zero-variance residual. Since $\mathrm{E}[W_n] = 0$, independence gives $\mathrm{E}[W_n \mid Y_n] = 0$. Taking the conditional expectation in (10), using Theorem 2.2, therefore yields

$$\boxed{g_n(y_0, \cdots, y_{n-1}) = \frac{c}{1 + (n-1)c} \sum_{j=0}^{n-1} y_j.} \tag{11}$$

This is the Gaussian regression theorem, proved in Theorem 2.5 from the Gaussian characteristic-function and conditional-expectation theorems. The nonsingular vector Y_n has a strictly positive density everywhere, so Lemma 2.6 establishes uniqueness of this continuous version even though a prescribed history has probability zero.

For the variance, put $S_n = \sum_{j=0}^{n-1} X_j$. Covariance bilinearity from Theorem 2.1 gives $\mathrm{Var}(S_n) = nd_n$ and $\mathrm{Cov}(X_n, S_n) = nc$. With $\alpha = c/d_n$,

$$\mathrm{Var}(W_n) = 1 + \alpha^2 n d_n - 2\alpha n c = 1 - \frac{nc^2}{d_n}.$$

Independence of W_n and Y_n, established above, then gives

$$\begin{aligned} \mathrm{Var}(X_n \mid Y_n) &= \mathrm{Var}(W_n) = 1 - \frac{nc^2}{1 + (n-1)c} \\ &= \frac{(1-c)(1+nc)}{1 + (n-1)c}. \end{aligned} \tag{12}$$

□

This nonnegative conditional variance describes actual test outcomes, not randomness in the recursively specified numbers e_n.

5 Solving the recursion and taking the average

Theorem 5.1 (One fixed starting score)**.** *Under the stated Gaussian model and continuous-version convention, for a fixed starting score a and $-1/k \le c < 1$, every subsequent recursive prediction equals ca, and the mean including the starting value is $B_k = [1 + kc]a/(k+1)$.*

Proof. Inserting the prescribed history into (11) gives

$$e_n = \frac{c}{1 + (n-1)c} \left(a + \sum_{j=1}^{n-1} e_j \right), \qquad 1 \le n \le k, \tag{13}$$

where the sum is empty at $n = 1$. In particular, $e_1 = ca$. Suppose $e_j = ca$ for all $1 \le j < n$. Then

$$e_n = \frac{c}{1 + (n-1)c}[a + (n-1)ca] = \frac{ca[1 + (n-1)c]}{1 + (n-1)c} = ca.$$

The base case and induction step satisfy Theorem 2.7 [5, Section 10.1]; hence

$$\boxed{e_1 = e_2 = \cdots = e_k = ca.} \tag{14}$$

Consequently the requested average is

$$\boxed{B_k = \frac{a + \sum_{n=1}^{k} e_n}{k+1} = \frac{1 + kc}{k+1}\, a = \left(c + \frac{1-c}{k+1} \right) a.} \tag{15}$$

Unlike the individual subsequent predictions, the requested average depends on the number of values included. With $\lambda_k = (1 + kc)/(k+1)$, it is $B_k = \lambda_k a$. Once a, c, k are fixed, this average is deterministic.

An additional expectation changes nothing: $\mathrm{E}[B_k] = \lambda_k a$. If $N = k+1$ denotes the *total* number of values, the same result is

$$\boxed{B_{N-1} = \frac{1 + (N-1)c}{N}\, a, \qquad N \geq 2.} \tag{16}$$

For the single-value case $N = 1$, the average is simply a, without any subsequent prediction or inter-test correlation being needed.

There is also a partial-sum derivation. Define

$$T_0 = a, \qquad T_n = a + \sum_{j=1}^{n} e_j.$$

Equation (13) implies

$$T_n = \frac{1 + nc}{1 + (n-1)c} T_{n-1}.$$

Multiplying these relations telescopes, without dividing by c or by a:

$$T_n = a \prod_{j=1}^{n} \frac{1 + jc}{1 + (j-1)c} = a(1 + nc).$$

Dividing $T_k = a + \sum_{j=1}^{k} e_j = a(1 + kc)$ by $k + 1$ again gives (15). This proof includes $a = 0$ and $c = 0$. □

For example, if $a = 2$ and $c = 0.6$, then

$$e_1 = 1.2, \qquad e_2 = \frac{0.6}{1.6}(2 + 1.2) = 1.2, \qquad e_3 = \frac{0.6}{2.2}(2 + 1.2 + 1.2) = 1.2.$$

Every subsequent prediction remains 1.2. Including the starting score, the average after three predictions is instead

$$B_3 = \frac{2 + 1.2 + 1.2 + 1.2}{4} = \frac{1 + 3(0.6)}{4}\, 2 = 1.4.$$

These substitutions and all displayed decimal rounding are certified in Section 11.

6 Standard deviation of the mean

6.1 The specified recursive average with fixed starting score

The numbers $a, e_1, \ldots, e_k$ and their average B_k are fixed once a, c, k are specified. The constant-variable case of Theorem 2.1, or the variance definition itself, proves

$$\mathrm{Var}(B_k) = \mathrm{E}[(B_k - \mathrm{E}[B_k])^2] = 0, \qquad \boxed{\mathrm{SD}(B_k) = 0.} \tag{17}$$

The actual test-score mean also includes the first test. Its population and conditional standard deviations are derived next.

6.2 Population standard deviation of the actual test-score mean

Proposition 6.1 (Population spread of the arithmetic mean)**.** *For centered, unit-variance scores with common covariance c,* $\mathrm{Var}(C_k) = [1 + kc]/(k + 1)$ *and* $\mathrm{SD}(C_k) = \sqrt{[1 + kc]/(k + 1)}$*, without requiring Gaussianity.*

Proof. Define the random mean of all $k + 1$ scores by

$$C_k = \frac{1}{k+1} \sum_{j=0}^{k} X_j. \tag{18}$$

Its population mean is zero. There are $k + 1$ variance terms, each equal to 1, and $k(k + 1)/2$ distinct covariance pairs, each equal to c. Consequently, by Theorem 2.1 [2, summary Section 4.2],

$$\begin{aligned} \mathrm{Var}(C_k) &= \frac{1}{(k+1)^2} \left(\sum_{j=0}^{k} \mathrm{Var}(X_j) + 2 \sum_{0 \le i < j \le k} \mathrm{Cov}(X_i, X_j) \right) \\ &= \frac{(k+1) + k(k+1)c}{(k+1)^2} = \frac{1 + kc}{k+1} = \lambda_k. \end{aligned} \tag{19}$$

Taking the nonnegative square root gives

$$\boxed{\mathrm{SD}(C_k) = \sqrt{\frac{1 + kc}{k+1}} = \sqrt{c + \frac{1 - c}{k+1}}.} \tag{20}$$

For $N = k + 1$ total tests, this is

$$\boxed{\mathrm{SD}(C_{N-1}) = \sqrt{\frac{1 + (N-1)c}{N}}.} \tag{21}$$

The population variance and standard deviation require only the specified finite variances and covariances; joint Gaussianity is not needed for (19)–(21). □

6.3 Conditional standard deviation after observing the first score

Proposition 6.2 (Spread after one observed score)**.** *Under the Gaussian model, the actual mean conditional on $X_0 = a$ has mean B_k, variance $\lambda_k(1 - \lambda_k)$, and standard deviation $\sqrt{k(1 - c)(1 + kc)}/(k + 1)$.*

Proof. Return to the joint Gaussian model and fix $X_0 = a$. For $1 \le i \le k$, write $U_i = X_i - cX_0$. The vector of residuals is jointly Gaussian with X_0, has zero covariance with X_0, and is therefore independent of it by Theorem 2.4. Covariance bilinearity (Theorem 2.1) gives

$$\begin{aligned} \mathrm{Var}(U_i) &= 1 - c^2, \\ \mathrm{Cov}(U_i, U_j) &= c - c^2 \qquad (i \neq j). \end{aligned}$$

Thus each future score has conditional mean ca, conditional variance $1 - c^2$, and each distinct future pair has conditional covariance $c - c^2$. Including the fixed first score gives

$$\mathrm{E}[C_k \mid X_0 = a] = \frac{a + kca}{k+1} = \lambda_k a = B_k. \tag{22}$$

The starting score contributes no variance after it is fixed, so

$$\begin{aligned}\operatorname{Var}(C_k \mid X_0 = a) &= \frac{k(1-c^2) + k(k-1)(c-c^2)}{(k+1)^2} \\ &= \frac{k(1-c)(1+kc)}{(k+1)^2} = \lambda_k(1-\lambda_k).\end{aligned} \tag{23}$$

Therefore

$$\boxed{\operatorname{SD}(C_k \mid X_0 = a) = \frac{\sqrt{k(1-c)(1+kc)}}{k+1}.} \tag{24}$$

This conditional standard deviation does not depend on the numerical value of a under the Gaussian model. The value of a determines the conditional mean. In terms of the total count $N = k+1$,

$$\operatorname{SD}(C_{N-1} \mid X_0 = a) = \frac{\sqrt{(N-1)(1-c)[1+(N-1)c]}}{N}. \tag{25}$$

These are Theorem 2.5 and the linearity of conditional expectation in Theorem 2.2, specialized to the first score. □

For the example $a = 2, c = 0.6, k = 3$, the four-score mean satisfies

$$\mathrm{E}[C_3 \mid X_0 = 2] = 1.4, \qquad \operatorname{SD}(C_3) = \sqrt{0.7} \approx 0.836660,$$

$$\operatorname{SD}(C_3 \mid X_0 = 2) = \sqrt{0.21} \approx 0.458258, \qquad \operatorname{SD}(B_3) = 0.$$

The population spread, the uncertainty remaining after the first score is observed, and the variation of the fixed recursive mean are different quantities. The exact values underlying these decimals are in Section 11.

6.4 Predictors based on actual earlier outcomes

The random predictors

$$P_n = \mathrm{E}[X_n \mid X_0, \dots, X_{n-1}]$$

use actual prior outcomes and are not the deterministic values e_n. The tower property (Theorem 2.2, with $\sigma(X_0) \subseteq \sigma(X_0, \dots, X_{n-1})$) proves $\mathrm{E}[P_n \mid X_0] = \mathrm{E}[X_n \mid X_0] = cX_0$, while its unconditional implication $\mathrm{E}[P_n] = 0$ holds without Gaussianity whenever the scores are integrable and have mean zero. Applying a conditional mean function at a fixed, recursively chosen history is a different operation.

6.5 The subsequent predictions considered separately

For comparison, Theorem 5.1 proves that averaging just the k subsequent predictions gives

$$\frac{1}{k}\sum_{n=1}^{k} e_n = ca. \tag{26}$$

The requested average B_k includes a as well and therefore has denominator $k+1$, as in (15).

6.6 Population mean zero and the fixed starting score

If one randomizes the starting observation according to its standardized test distribution, the recursively filled average becomes $\lambda_k X_0$. The affine-moment identities in Theorem 2.1 prove

$$\mathrm{E}[\lambda_k X_0] = 0, \qquad \mathrm{SD}(\lambda_k X_0) = \lambda_k, \tag{27}$$

where $\lambda_k \geq 0$ throughout the admissible correlation domain. These are population statements obtained by varying the starting score. In particular, the standard deviation λ_k of the recursively filled mean across random starting scores generally differs from the standard deviation $\sqrt{\lambda_k}$ of the actual test-score mean. The law of total variance accounts for the difference; its proof is included in Theorem 2.2 [2, summary Section 4.3]:

$$\begin{aligned}\mathrm{Var}(C_k) &= \mathrm{Var}(\mathrm{E}[C_k \mid X_0]) + \mathrm{E}[\mathrm{Var}(C_k \mid X_0)] \\ &= \lambda_k^2 + \lambda_k(1 - \lambda_k) = \lambda_k.\end{aligned}$$

For the fixed starting score in this problem, the answer is $B_k = \lambda_k a$, which need not be zero. Every original test still has mean 0 and standard deviation 1; the arithmetic average has its own distribution and is not rescaled to unit standard deviation.

7 Two or three initial scores equal to the starting value

7.1 General setup with several fixed starting scores

Let $r \geq 1$ be the number of starting scores and keep $k \geq 1$ for the number of subsequent predictions. There are now $N = r + k$ jointly Gaussian scores $X_0, \ldots, X_{N-1}$, each with population mean 0, standard deviation 1, and pairwise correlation c. Fix the first r scores at the same value:

$$X_0 = X_1 = \cdots = X_{r-1} = a. \tag{28}$$

These are r distinct tests with equal observed scores. Define

$$d_r = 1 + (r-1)c, \qquad d_N = 1 + (N-1)c.$$

The covariance-matrix eigenvalues give the domain

$$-\frac{1}{N-1} \leq c \leq 1. \tag{29}$$

This is Proposition 3.1 with k there replaced by $N - 1$; the Gaussian existence proof therefore applies to this larger vector as well. For $c < 1$, all conditioning prefixes are nonsingular. In particular,

$$d_r \geq 1 - \frac{r-1}{N-1} = \frac{k}{N-1} > 0.$$

As before, evaluation at an exact history means the continuous Gaussian conditional-mean version. This specifies the equal-score histories even though their point probabilities are zero.

7.2 Every subsequent prediction and the inclusive average

Theorem 7.1 (Several equal fixed starting scores)**.** *Let $r \geq 1$, $k \geq 1$, $N = r + k$, and $-1/(N-1) \leq c < 1$. Given r initial scores all fixed at a, every subsequent recursive prediction is rca/d_r, and the inclusive mean is $B_{r,k} = rd_N a/(Nd_r)$.*

Proof. Let $e_j^{(r)}$ be the recursive prediction of X_{r+j-1}, using the r fixed initial scores and the previous $j-1$ predictions. Apply Theorem 4.1 with its k replaced by $N-1$, and its n replaced by $r+j-1$. Since $1 \leq r+j-1 \leq N-1$, all its hypotheses hold. Equation (11) therefore gives

$$e_j^{(r)} = \frac{c}{d_r + (j-1)c}\left(ra + \sum_{\ell=1}^{j-1} e_\ell^{(r)}\right), \qquad 1 \leq j \leq k. \tag{30}$$

The first prediction is $b_r = rca/d_r$. If the previous predictions all equal b_r, then

$$ra + \sum_{\ell=1}^{j-1} e_\ell^{(r)} = ra + (j-1)\frac{rca}{d_r} = \frac{ra}{d_r}[d_r + (j-1)c].$$

Substituting into (30) and cancelling the positive denominator establishes the induction step. Theorem 2.7 then proves

$$\boxed{e_j^{(r)} = b_r = \frac{rc}{1 + (r-1)c}\,a, \qquad 1 \leq j \leq k.} \tag{31}$$

The requested average includes all r starting values and all k subsequent predictions, so

$$\begin{aligned} B_{r,k} &= \frac{ra + \sum_{j=1}^{k} e_j^{(r)}}{r+k} = \frac{ra + k(rca/d_r)}{N} \\ &= \boxed{\frac{r\,[1 + (r+k-1)c]}{(r+k)[1 + (r-1)c]}\,a.} \end{aligned} \tag{32}$$

□

Putting $r = 1$ recovers (15). Holding all starting scores, c, and the counts fixed makes $B_{r,k}$ deterministic, with $\mathrm{SD}(B_{r,k}) = 0$.

7.3 Population and conditional standard deviations

Theorem 7.2 (Spread with several observed starting scores)**.** *Under the preceding assumptions, the actual mean $C_{r,k}$ has population variance d_N/N. Conditional on the initial block being $a\mathbf{1}_r$, its mean is $B_{r,k}$ and its variance is $k(1-c)d_N/(N^2 d_r)$. The population-variance statement requires only the covariance structure; the conditional statement uses joint Gaussianity.*

Proof. Let the actual mean of the $N = r + k$ test outcomes be

$$C_{r,k} = \frac{1}{N}\sum_{i=0}^{N-1} X_i.$$

Theorem 2.1 and the covariance count in Proposition 6.1 give

$$\boxed{\mathrm{SD}(C_{r,k}) = \sqrt{\frac{1+(r+k-1)c}{r+k}} = \sqrt{\frac{d_N}{N}}.} \tag{33}$$

This population formula does not require Gaussianity. The following conditional calculation does use the joint Gaussian model.

Let $Z = (X_0, \dots, X_{r-1})^{\mathsf{T}}$. Since $\mathrm{Cov}(Z) = (1-c)I_r + c\mathbf{1}_r\mathbf{1}_r^{\mathsf{T}}$, the projection of each future score onto the starting block is $(c/d_r)\sum_{i=0}^{r-1} X_i$, by Theorem 2.5 and the inverse identity (8) with $n = r$. Here the battery parameter in Theorem 4.1 is $N-1$, so $r \le N-1$ is within its conditioning-prefix range. Define the future residuals

$$V_j = X_{r+j-1} - \frac{c}{d_r}\sum_{i=0}^{r-1} X_i, \qquad 1 \le j \le k.$$

For each initial coordinate X_ℓ, covariance bilinearity gives $\mathrm{Cov}(V_j, X_\ell) = c - (c/d_r)d_r = 0$. Joint Gaussianity therefore makes the residual vector independent of Z, by Theorem 2.4. Put $S_r = \sum_{i=0}^{r-1} X_i$ and $\beta = c/d_r$. Theorem 2.1 gives $\mathrm{Var}(S_r) = rd_r$ and $\mathrm{Cov}(X_{r+j-1}, S_r) = rc$. Expanding the residual covariance entries yields

$$\begin{aligned}
\mathrm{Var}(V_j) &= 1 + \beta^2 r d_r - 2\beta r c = 1 - \frac{rc^2}{d_r}, \\
\mathrm{Cov}(V_i, V_j) &= c + \beta^2 r d_r - 2\beta r c = c - \frac{rc^2}{d_r} = \frac{c(1-c)}{d_r} \qquad (i \ne j).
\end{aligned}$$

Equivalently, the conditional covariance matrix of the future scores is

$$(1-c)I_k + \frac{c(1-c)}{d_r}\mathbf{1}_k\mathbf{1}_k^{\mathsf{T}}.$$

Conditioning on $Z = a\mathbf{1}_r$ fixes all r starting terms. The future conditional means are all rca/d_r, so $\mathrm{E}[C_{r,k} \mid Z = a\mathbf{1}_r] = B_{r,k}$. Summing the residual covariance terms by Theorem 2.1 gives

$$\begin{aligned}
\mathrm{Var}(C_{r,k} \mid Z = a\mathbf{1}_r) &= \frac{1}{N^2}\left[k(1-c) + \frac{k^2 c(1-c)}{d_r}\right] \\
&= \frac{k(1-c)(d_r + kc)}{N^2 d_r} = \frac{k(1-c)d_N}{N^2 d_r}.
\end{aligned} \tag{34}$$

Thus the conditional standard deviation of the actual mean is

$$\boxed{\mathrm{SD}(C_{r,k} \mid Z = a\mathbf{1}_r) = \frac{1}{r+k}\sqrt{\frac{k(1-c)[1+(r+k-1)c]}{1+(r-1)c}}.} \tag{35}$$

□

It is independent of the numerical value of a. This is the spread of the actual mean after the starting block is observed; the fixed recursive mean itself has standard deviation zero. An across-population calculation with random initial scores uses a random starting *block*. The single random-start interpretation in (27) does not specify a distribution for several starting scores constrained to share one random value.

7.4 Special case: two starting scores equal to a

Corollary 7.3 (Two starting scores)**.** *Set $r = 2$. The initial values are a, a, followed by k predictions, so $N = k+2$ and $-1/(k+1) \le c \le 1$. Every prediction and the inclusive average are*

$$e_j^{(2)} = \boxed{\frac{2ca}{1+c}}, \qquad 1 \le j \le k, \tag{36}$$

$$B_{2,k} = \frac{2a + \sum_{j=1}^{k} e_j^{(2)}}{k+2} = \boxed{\frac{2[1+(k+1)c]}{(k+2)(1+c)}\, a.} \tag{37}$$

The actual mean has population and conditional standard deviations

$$\mathrm{SD}(C_{2,k}) = \boxed{\sqrt{\frac{1+(k+1)c}{k+2}}}, \tag{38}$$

$$\mathrm{SD}(C_{2,k} \mid X_0 = X_1 = a) = \boxed{\frac{1}{k+2}\sqrt{\frac{k(1-c)[1+(k+1)c]}{1+c}}.} \tag{39}$$

Proof. Substitute $r = 2$, $d_r = 1 + c$, and $N = k + 2$ into Theorems 7.1 and 7.2. Proposition 3.1 gives the stated correlation domain. The endpoint $c = 1$ follows from Proposition 7.5. □

7.5 Special case: three starting scores equal to a

Corollary 7.4 (Three starting scores)**.** *Set $r = 3$. The initial values are a, a, a, followed by k predictions, so $N = k + 3$ and $-1/(k+2) \le c \le 1$. The predictions and inclusive average are*

$$e_j^{(3)} = \boxed{\frac{3ca}{1+2c}}, \qquad 1 \le j \le k, \tag{40}$$

$$B_{3,k} = \frac{3a + \sum_{j=1}^{k} e_j^{(3)}}{k+3} = \boxed{\frac{3[1+(k+2)c]}{(k+3)(1+2c)}\, a.} \tag{41}$$

The actual mean has standard deviations

$$\mathrm{SD}(C_{3,k}) = \boxed{\sqrt{\frac{1+(k+2)c}{k+3}}}, \tag{42}$$

$$\mathrm{SD}(C_{3,k} \mid X_0 = X_1 = X_2 = a) = \boxed{\frac{1}{k+3}\sqrt{\frac{k(1-c)[1+(k+2)c]}{1+2c}}.} \tag{43}$$

Proof. Substitute $r = 3$, $d_r = 1 + 2c$, and $N = k + 3$ into Theorems 7.1 and 7.2. Proposition 3.1 gives the stated domain, and Proposition 7.5 supplies $c = 1$. □

For both special cases, the standard deviation of the fixed recursive average $B_{r,k}$ is 0, by Theorem 2.1.

7.6 Checks, endpoints and an example

Proposition 7.5 (Boundary values and compatible histories)**.** *The preceding formulas extend to $c = 1$ on compatible equal-score histories. At $c = 0$ they give independent future scores; at $c = -1/(N-1)$ the inclusive mean and both actual-mean SDs are zero. If $k = 0$, the fixed list averages to a, with conditional SD zero, provided its history belongs to the model support.*

Proof. At $c = 0$, every subsequent prediction is zero, $B_{r,k} = ra/(r+k)$, and the actual mean has population SD $1/\sqrt{r+k}$ and conditional SD $\sqrt{k}/(r+k)$. These values are direct substitutions into Theorems 7.1 and 7.2; independence follows from Theorem 2.4. At $c = 1$, all test scores agree almost surely. The equal starting history is compatible with this support; all predictions and their inclusive mean equal a. The population SD is 1, and the conditional SD is 0, establishing the endpoint directly. Indeed, $\mathrm{Var}(X_i - X_j) = 2(1-c) = 0$. The zero-variance consequence of Chebyshev's inequality in Theorem 2.1 gives equality of every pair almost surely; a finite union of null events gives simultaneous equality. On the diagonal support the continuous conditional law therefore sets every subsequent score to a.

At $c = -1/(N-1)$, Theorem 2.1 gives $\mathrm{Var}(\sum_{i=0}^{N-1} X_i) = Nd_N = 0$. Its zero-variance conclusion therefore makes the total score zero almost surely. The initial block remains nonsingular because $k \geq 1$, and

$$e_j^{(r)} = -\frac{ra}{k}, \qquad ra + \sum_{j=1}^{k} e_j^{(r)} = 0.$$

Hence $B_{r,k} = 0$, and both actual-mean standard deviations are zero. If there are no subsequent predictions, the fixed list of r values a simply averages to a, with conditional SD zero, provided the starting history is compatible with the model. The general derivation above assumes $k \geq 1$; it does not divide by a singular initial-block eigenvalue when $k = 0$. For $r \geq 2$ and $k = 0$, the lower endpoint $c = -1/(r-1)$ permits equal starting scores only when $a = 0$: the same zero-variance argument forces their total ra to vanish. □

For $a = 2$, $c = 0.6$, and $k = 3$ subsequent predictions, the two special cases give the following values, rounded to six decimal places:

Quantity	Two initial scores	Three initial scores
Each subsequent prediction	1.500000	1.636364
Mean including all starting scores	1.700000	1.818182
Population SD of the actual mean	0.824621	0.816497
Conditional SD of the actual mean	0.319374	0.246183
SD of the fixed recursive mean	0	0

Proof of the numerical table. Substitute $a = 2$, $c = 3/5$, and $k = 3$ into Corollaries 7.3 and 7.4. Section 11 records the exact fractions, variances and rounding bounds, so no floating-point accuracy assumption is needed for the printed six-decimal entries. □

8 Worked examples starting at IQ 150

Corollary 8.1 (Affine reporting of the proved results)**.** *With the definition $Q_i = 100 + 15X_i$, the IQ-150 starting score is $a = 10/3$. Equations (46), (48), (49) and (50) give the subsequent IQ predictions, inclusive means and actual-mean SDs. The fixed recursive mean still has SD zero.*

8.1 Convert the reporting scale to the standardized model

Proof. The model scores X_i continue to have population mean 0 and standard deviation 1. To report them on an IQ scale with mean 100 and standard deviation 15, define

$$Q_i = 100 + 15X_i. \tag{44}$$

An initial observed IQ of 150 therefore corresponds to

$$a = \frac{150 - 100}{15} = \frac{10}{3}. \tag{45}$$

Theorem 2.1 gives $\mathrm{E}[Q_i] = 100$, $\mathrm{Var}(Q_i) = 225$, and $\mathrm{Cov}(Q_i, Q_j) = 225c$, so the correlation is $225c/(15 \cdot 15) = c$. The mean and SD of the standardized model are unchanged; Q_i is its reporting scale. In the examples below, $r \in \{1, 2, 3\}$ is the number of initial IQ-150 scores, $k \in \{1, 2, 10\}$ is the number of subsequent predictions, and every initial score is included in the $N = r + k$ values being averaged. Use the selected correlation $c = 0.6 = 3/5$ throughout this section.

8.2 Predictions, means and standard deviations in IQ points

Put $d_r = 1 + (r-1)c$ and $d_N = 1 + (N-1)c$, as above. Conditional linearity from Theorem 2.2, together with Theorem 7.1, gives the IQ prediction for every subsequent score:

$$q_r = 100 + 15\frac{rca}{d_r} = 100 + 50\frac{rc}{d_r}. \tag{46}$$

At $c = 0.6$, these predictions are

$$q_1 = 130, \qquad q_2 = 137.5, \qquad q_3 = \frac{1550}{11} \approx 140.909091. \tag{47}$$

They depend on the number of fixed starting scores r, and are constant across all k subsequent recursive predictions within a case.

The recursive average in IQ points, also equal to the conditional expected value of the actual IQ-score mean, is

$$m^{\mathrm{IQ}}_{r,k} = 100 + 15B_{r,k} = 100 + 50\frac{rd_N}{Nd_r} = \frac{150r + kq_r}{r + k}. \tag{48}$$

Conditional linearity in Theorem 2.2 proves $\mathrm{E}[100 + 15W \mid Z] = 100 + 15\,\mathrm{E}[W \mid Z]$. Subtracting this conditional mean from $100 + 15W$ leaves $15(W - \mathrm{E}[W \mid Z])$. Squaring and taking conditional expectation therefore proves

$$\mathrm{Var}(100 + 15W \mid Z) = 225\,\mathrm{Var}(W \mid Z), \qquad \mathrm{SD}(100 + 15W \mid Z) = 15\,\mathrm{SD}(W \mid Z).$$

Applying Theorem 7.2 therefore gives the actual IQ-score mean's population and conditional standard deviations

$$\sigma^{\mathrm{IQ}}_{\mathrm{pop},r,k} = 15\sqrt{\frac{d_N}{N}}, \tag{49}$$

$$\sigma^{\mathrm{IQ}}_{\mathrm{cond},r,k} = \frac{15}{N}\sqrt{\frac{k(1-c)d_N}{d_r}}. \tag{50}$$

The conditional SD is calculated after all r initial IQ-150 scores have been observed. The fixed recursively filled average itself has SD 0. □

8.3 The nine requested examples

The following table reports the inclusive mean and both actual-mean SDs in IQ points, rounded to two decimal places. Here $N = r + k$ includes every starting score and every subsequent prediction. The mean is $m^{\mathrm{IQ}}_{r,k}$ from (48); the SD columns use (49) and (50).

Initial scores r	Subsequent predictions k	Total N	Mean IQ	Population SD	Conditional SD
1	1	2	140.00	13.42	6.00
1	2	3	136.67	12.85	6.63
1	10	11	131.82	11.97	7.22
2	1	3	145.83	12.85	3.71
2	2	4	143.75	12.55	4.44
2	10	12	139.58	11.94	5.45
3	1	4	147.73	12.55	2.68
3	2	5	146.36	12.37	3.34
3	10	13	143.01	11.91	4.46

For example, with two starting scores of 150 and ten subsequent predictions, each subsequent IQ prediction is 137.5. The inclusive mean is therefore

$$\frac{2(150) + 10(137.5)}{12} = \frac{1675}{12} \approx 139.583333.$$

In this row, the population variance of the actual IQ mean is $285/2$, and its conditional variance is $475/16$. Their square roots give approximately 11.937336 and 5.448624 IQ points, respectively. The fixed recursive mean in the same row remains a deterministic number.

Proof of the numerical IQ examples. All nine rows follow by substituting $a = 10/3$, $c = 3/5$, $r \in \{1, 2, 3\}$, and $k \in \{1, 2, 10\}$ into Corollary 8.1. Section 11 provides their exact rational means and variances and strict integer inequalities certifying every displayed two-decimal rounding. The worked row uses the same exact fractions, with its six-decimal bounds certified there as well. □

9 Boundary cases with one starting score

Proof of the one-start boundary statements. These are the $r = 1$ cases of Proposition 7.5. For clarity, the individual consequences are spelled out below.

At $c = 0$, joint Gaussianity makes the scores independent, and every subsequent prediction is zero. Including the starting score gives $B_k = a/(k + 1)$. The population and conditional standard deviations of the actual mean are then $1/\sqrt{k + 1}$ and $\sqrt{k}/(k + 1)$, respectively. At $c = 1$, $\mathrm{Var}(X_i - X_j) = 2(1 - c) = 0$, so all scores agree almost surely. Given the starting score a, every compatible recursive prediction is $a = ca$, and $B_k = a$. This establishes the result at $c = 1$ without using the singular inverse in (8). The population standard deviation of C_k is 1, while its conditional standard deviation after observing $X_0 = a$ is 0.

At the other endpoint, $c = -1/k$, the full vector satisfies

$$\sum_{j=0}^{k} X_j = 0 \quad \text{almost surely,}$$

because that sum has mean zero and variance $(k+1)(1+kc) = 0$. Every conditioning prefix remains nonsingular, as shown above, so the derivation of (15) applies. Its prescribed history is compatible with the support, since $a + kca = 0$, so $B_k = 0$. The final conditional variance (12) is zero. Both the population and conditional standard deviations of C_k are also zero at this endpoint. The fixed recursive mean has standard deviation zero at every admissible correlation. For the single-value case $N = 1$, the actual score has population standard deviation 1, and fixing its value leaves conditional standard deviation 0. The almost-sure conclusions follow from the zero-variance part of Theorem 2.1; the single-value statements follow directly from $C_0 = X_0$ and the constant-variable variance definition. □

10 Why correlation alone is insufficient

Proposition 10.1 (Conditional means are not determined by the marginals)**.** *Two pairs can have the same one-dimensional marginal distributions, mean zero, variance one, and correlation zero, yet have different conditional means and different recursive averages at a starting value of positive probability.*

Proof. The original description specifies only marginal means, marginal variances and pairwise correlations. Those quantities do not determine the conditional mean functions. A counterexample already exists with one subsequent test. Let

$$P(X_0 = -\sqrt{2}) = P(X_0 = \sqrt{2}) = \tfrac{1}{4}, \qquad P(X_0 = 0) = \tfrac{1}{2}, \qquad X_1 = X_0^2 - 1.$$

Symmetry and direct calculation give

$$\mathrm{E}[X_0] = 0, \quad \mathrm{E}[X_0^2] = 1, \quad \mathrm{E}[X_0^3] = 0, \quad \mathrm{E}[X_0^4] = 2.$$

Consequently

$$\begin{aligned} \mathrm{E}[X_1] &= \mathrm{E}[X_0^2] - 1 = 0, \\ \mathrm{Var}(X_1) &= \mathrm{E}[(X_0^2 - 1)^2] = 2 - 2 + 1 = 1, \\ \mathrm{Cov}(X_0, X_1) &= \mathrm{E}[X_0^3 - X_0] = 0. \end{aligned}$$

Both scores are standardized and $c = 0$, yet at the fixed starting value $a = 0$, an event with positive probability,

$$e_1 = \mathrm{E}[X_1 \mid X_0 = 0] = -1 \neq 0 = ca.$$

The average including the starting value is therefore

$$B_1 = \frac{0 + (-1)}{2} = -\frac{1}{2},$$

whereas (15) would give zero under the Gaussian model. Therefore (15) is a result under the stated joint Gaussian model, or another model with the same conditional-mean functions; it does not follow from equicorrelation alone.

For an explicit comparison with identical marginal distributions, let X_1' be independent of X_0, with probabilities $1/2$ at each of -1 and 1. The variable $X_1 = X_0^2 - 1$ has exactly this same marginal distribution. By independence and Theorem 2.1, (X_0, X_1') also has correlation zero and the same marginal means and variances as (X_0, X_1). But Theorem 2.2 gives $\mathrm{E}[X_1' \mid X_0 = 0] = 0$, whereas the first pair gives -1. The conditioning event has probability $1/2$ in both models, so this difference cannot be attributed to a choice of versions on a null history. Their inclusive recursive averages at $a = 0$ are respectively 0 and $-1/2$. This proves the proposition. □

11 Exact arithmetic certificates for numerical examples

The numerical examples are substitutions into the proved mean and variance formulas. The exact fractions below, together with the rounding lemma, provide arithmetic proofs of the displayed decimals. They do not depend on floating-point output, simulation, or the software-verification results.

11.1 Exact values behind the IQ-150 examples

Corollary 11.1 (Exact IQ means and variances)**.** *Under the joint Gaussian model, let $r, k \geq 1$ be integers and set $c = 3/5$, $a = 10/3$, and $N = r + k$. On the IQ reporting scale $Q = 100 + 15X$, write V_{pop} and V_{cond} for the population variance and the variance conditional on all r initial IQ scores being 150, respectively, of the actual IQ-score mean. Then*

$$m_{r,k}^{\mathrm{IQ}} = 100 + \frac{50r(3N+2)}{N(3r+2)}, \tag{51}$$

$$V_{\mathrm{pop}} = \frac{45(3N+2)}{N}, \tag{52}$$

$$V_{\mathrm{cond}} = \frac{90k(3N+2)}{N^2(3r+2)}. \tag{53}$$

Proof. The definitions give

$$d_r = 1 + (r-1)\frac{3}{5} = \frac{3r+2}{5}, \qquad d_N = 1 + (N-1)\frac{3}{5} = \frac{3N+2}{5}.$$

Substitution into (48) gives $100 + 50rd_N/(Nd_r)$, which is (51). Squaring (49) gives $225d_N/N$, hence (52). Squaring (50) gives

$$\frac{225k(1-3/5)d_N}{N^2 d_r} = \frac{90k(3N+2)}{N^2(3r+2)},$$

which is (53). The affine moment identities in Theorem 2.1 justify the factor $15^2 = 225$ for variances and the additive 100 for means. □

For the nine parameter pairs in Section 8, the resulting exact quantities are as follows. The SDs are the nonnegative square roots of the two variance columns.

r	k	N	Exact IQ mean	V_{pop}	V_{cond}
1	1	2	140	180	36
1	2	3	410/3	165	44
1	10	11	1450/11	1575/11	6300/121
2	1	3	875/6	165	55/4
2	2	4	575/4	315/2	315/16
2	10	12	1675/12	285/2	475/16
3	1	4	1625/11	315/2	315/44
3	2	5	1610/11	153	612/55
3	10	13	20450/143	1845/13	36900/1859

For example, the last row has $r = 3$, $k = 10$, and $N = 13$. Direct integer substitution yields

$$100 + \frac{50 \cdot 3 \cdot 41}{13 \cdot 11} = \frac{20450}{143}, \qquad \frac{45 \cdot 41}{13} = \frac{1845}{13}, \qquad \frac{90 \cdot 10 \cdot 41}{13^2 \cdot 11} = \frac{36900}{1859}.$$

The other rows follow by the same substitutions; all quantities in this table are exact rational numbers, rather than decimal approximations.

11.2 A proof of the decimal rounding rule

Lemma 11.2 (Exact rounding test)**.** *Let $T = 10^p$, where p is a positive integer, and let t be an integer. Rounding a real number to p decimal places means choosing a nearest point of the grid $\{j/T : j \in \mathbb{Z}\}$.*

1. *For $x = P/Q$, with integers P, Q and $Q > 0$, define*

$$D_T = Q - 2|TP - tQ|. \tag{54}$$

   If $D_T > 0$, the unique nearest grid point to x is t/T.

2. *For a variance $v = P/Q \geq 0$, with integers P, Q and $Q > 0$, and an integer $t \geq 1$, define*

$$\begin{aligned} L_T &= 4T^2P - (2t-1)^2Q, \\ U_T &= (2t+1)^2Q - 4T^2P. \end{aligned} \tag{55}$$

   If $L_T > 0$ and $U_T > 0$, the unique nearest grid point to $\sqrt{v}$ is t/T.

All these strict inequalities exclude a halfway tie.

Proof. For the first part, $D_T > 0$ is equivalent to

$$|P/Q - t/T| = \frac{|TP - tQ|}{TQ} < \frac{1}{2T}.$$

Every other grid point is at distance at least $1/T$ from t/T, so the triangle inequality makes its distance from x strictly greater than $1/(2T)$. Thus t/T is the unique nearest grid point.

For the second part, divide the two positive integer differences by $4T^2Q$. They give

$$\left(\frac{2t-1}{2T}\right)^2 < v < \left(\frac{2t+1}{2T}\right)^2.$$

Both bounding numbers before squaring are positive. Squaring is strictly increasing on the nonnegative real numbers, so these inequalities imply

$$\frac{t - 1/2}{T} < \sqrt{v} < \frac{t + 1/2}{T}.$$

Therefore $|\sqrt{v} - t/T| < 1/(2T)$, and the grid-distance argument in the first part proves the result. This uses only the order properties of the real numbers and the definition of nearest-decimal rounding. □

11.3 Positive integer certificates for every IQ table row

For each row of the displayed IQ table, take $T = 100$. Use the reduced fraction P/Q in the exact-value table above. For the mean, t is 100 times its displayed two-decimal value. For each SD, t is 100 times its displayed value, while P/Q is its exact *variance*. The following table gives the integer differences from (54) and (55). In the SD formulas, $4T^2 = 40000$.

r	k	Mean D_{100}	L_{pop}	U_{pop}	L_{cond}	U_{cond}
1	1	1	1511	9225	2399	2401
1	2	1	239	10041	4375	929
1	10	7	9061	96275	47871	651025
2	1	2	239	10041	3676	8196
2	2	4	9838	10242	11696	45136
2	10	4	4462	14642	25264	44496
3	1	5	9838	10242	6100	88236
3	2	3	4271	5625	11105	135855
3	10	57	100907	22957	175221	6457691

Every listed integer is strictly positive. Lemma 11.2 therefore proves each of the nine mean roundings and all eighteen SD roundings in the IQ table. For example, in the $r = 3, k = 10$ row the displayed conditional SD is 4.46, so $t = 446$ and $v = 36900/1859$. The exact differences are

$$40000 \cdot 36900 - 891^2 \cdot 1859 = 175221 > 0,$$

$$893^2 \cdot 1859 - 40000 \cdot 36900 = 6457691 > 0.$$

Thus the true SD is strictly between 4.455 and 4.465, proving that its nearest two-decimal representation is 4.46.

11.4 Certificates for the other numerical examples

For the earlier standardized examples, substitute $a = 2$, $c = 3/5$, and $k = 3$ into (31), (32), (33), and (35). With $N = r + 3$, the exact quantities are

r	Each prediction b_r	Inclusive mean $B_{r,3}$	Population variance	Conditional variance
1	6/5	7/5	7/10	21/100
2	3/2	17/10	17/25	51/500
3	18/11	20/11	2/3	2/33

In particular, the terminating values 1.2, 1.4, 1.5, and 1.7, and their zero-padded forms, equal the corresponding fractions exactly. Each zero SD stated for a fixed recursive mean follows from Theorem 2.1, rather than a rounding approximation.

The nonterminating means and predictions used elsewhere have the following certificates. Here P/Q denotes the exact value itself, and each positive D_T is computed by (54).

Quantity	P/Q	T	Displayed value	D_T
b_3, standardized example	18/11	10^6	1.636364	3
$B_{3,3}$, standardized example	20/11	10^6	1.818182	7
q_3, IQ-150 prediction	1550/11	10^6	140.909091	9
q_3, IQ-150 prediction	1550/11	100	140.91	9
Worked IQ mean, $r = 2, k = 10$	1675/12	10^6	139.583333	4

The other IQ predictions, $q_1 = 130$ and $q_2 = 275/2 = 137.5$, are exact terminating values.

Finally, the following positive integer differences prove all displayed six-decimal SD values. In this table $T = 10^6$, P/Q is the exact variance, and L_T, U_T are given by (55).

SD quantity	Variance P/Q	Displayed SD	L_T	U_T
$r = 1$, population	7/10	0.836660	35242390	31690410
$r = 1$, conditional	21/100	0.458258	25477500	341128900
$r = 2$, population	17/25	0.824621	103097975	61826225
$r = 2$, conditional	51/500	0.319374	1134995500	142500500
$r = 3$, population	2/3	0.816497	1585853	18010075
$r = 3$, conditional	2/33	0.246183	31323575	33668737
Worked IQ, population	285/2	11.937336	169283518	21713858
Worked IQ, conditional	475/16	5.448624	125135856	572288016

Every entry in the last two certificate columns is strictly positive, so Lemma 11.2 proves the claimed six-decimal roundings. The exact fractions also establish the worked IQ-row variances 285/2 and 475/16. All decimal claims in the numerical examples are therefore either exact terminating fractions or consequences of these strict rational inequalities.

12 Map from claims to proofs and references

The following map covers the mathematical claims in this analysis. The reference theorems are stated with verified locators in Section 2; each project-specific deduction is proved in the location listed below. Model assumptions and definitions are inputs to those deductions, rather than empirical conclusions of a theorem.

Claim group	Proof in this report	Reference foundation
Means, variances, covariance sums, affine SDs and constant variables	Theorem 2.1; variance counts in Proposition 6.1	Expectation linearity, covariance definition, and Chebyshev's inequality [2, 3]
Admissible correlations and existence of the Gaussian model	Proposition 3.1; its square-root construction	Real spectral theorem and positive square root [1, Theorems 7.29 and 7.38]; Gaussian affine closure [4]
Gaussian independence, conditional means and conditional covariance	Theorems 2.4, 2.5 and 4.1	Gaussian characteristic function and multivariate inversion [3, 4]; the proof allows singular full vectors
Exact-history conditioning and matrix inverse	Lemma 2.6; explicit inverse multiplication in Theorem 4.1	Conditional-expectation uniqueness and positive Gaussian density [3, 4]; elementary matrix multiplication
One, two and three starting scores, subsequent predictions and inclusive means	Theorems 5.1 and 7.1; Corollaries 7.3 and 7.4	Gaussian regression and mathematical induction [4, 5]

Claim group	Proof in this report	Reference foundation
Population SD, conditional SD and SD zero for fixed means	Propositions 6.1 and 6.2; Theorem 7.2; (17)	Covariance bilinearity, Gaussian conditioning and the SD definition [2, 4]
Random predictors, random starts and total variance	Theorem 2.2 and its orthogonal-decomposition proof; (27)	Tower and pull-out theorems [3, Theorems 4.1.13–4.1.14]; law of total variance [2, summary Section 4.3]
Singular endpoints and compatible histories	Proposition 7.5 and its one-start specialization	Zero variance implies an almost-sure constant [3, Theorem 1.6.4]; Gaussian support and residual proof
IQ conversion, numerical means and SDs, and every printed rounding	Corollary 8.1; Section 11, including Lemma 11.2 and exact integer certificates	Affine expectation/variance identities [3]; proved mean/SD formulas and exact rational arithmetic
Failure of identification from moments or marginal laws	Proposition 10.1, by two explicit probability laws	Conditional expectation on a positive-probability atom and independence [3, Section 4.1]

Scope and provenance. This derivation and its AI review extend the repository's hypothetical Gaussian model. Existing Ada/SPARK and Python/Deal evidence concerns the specified software; it does not formally verify this text. No new empirical study or formal-prover campaign was performed. The accompanying README records build and visual checks. The repository's original material follows its existing license, subject to third-party rights. Source versions and access limits are also recorded in the accompanying `reference_notes.md`.

References

[1] Sheldon Axler. *Linear Algebra Done Right*. Springer, 4th edition, 2024. doi: 10.1007/978-3-031-41026-0. URL https://linear.axler.net/LADR4e.pdf. Theorem and page citations refer to the author's PDF dated 4 November 2026, at the linked URL. The fourth edition carries copyright 2024. Accessed 14 November 2026.

[2] Dimitri P. Bertsekas and John N. Tsitsiklis. *Introduction to Probability*. Athena Scientific, Belmont, Massachusetts, 2nd edition, 2008. ISBN 9781886529236. URL https://ocw.mit.edu/courses/res-6-012-introduction-to-probability-spring-2018/d973b10c2587781f86ca4f2aff49098f_MITRES_6_012S18_Textbook.pdf. The linked document is Selected Summary Material supplied by the authors through MIT OpenCourseWare. Citations use its summary pagination, especially pp. 10 and 31–32, not the full book's page numbers. Accessed 14 November 2026.

[3] Rick Durrett. *Probability: Theory and Examples*. Cambridge University Press, 5th edition, 2019. ISBN 9781108473682. doi: 10.1017/9781108591034. URL https://sites.math.duke.edu/~rtd/PTE/PTE5_011119.pdf. Theorem, equation and page citations refer to the author's Version 5, dated 11 January 2019, at the linked URL. Its pagination differs from the published edition. Accessed 14 November 2026.

[4] Robert G. Gallager. *Stochastic Processes: Theory for Applications*. Cambridge University Press, 2013. ISBN 9781107039759. doi: 10.1017/CBO9781139626514. URL https://www.rle.mit.edu/rgallager/documents/6.262lateweb3.pdf. Citations refer to the author's Chapter 3 draft, dated 2 December 2013, printed pp. 109–166, at the linked URL. The published chapter uses pp. 105–160. Accessed 14 November 2026.

[5] Richard Hammack. *Book of Proof*. Richard Hammack, Richmond, Virginia, 3rd edition, 2018. ISBN 9780989472128. URL https://richardhammack.github.io/BookOfProof/Main.pdf. Section and page citations refer to version 3.4, issued 5 February 2025, of the third edition, copyright 2018. Accessed 14 November 2026.